Kwabena Kingsley Kumah, Ph.D.

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Research Profile

Satellite precipitation scientist developing and evaluating retrievals and climate records where precipitation is difficult to observe: high latitudes, oceans, Antarctica, and data-sparse tropical regions. At the University of Arizona, contributes to NASA-supported GPCP research spanning AVHRR-based high-latitude retrieval, multi-reference ocean evaluation, and observation-constrained Antarctic snowfall assessment. Doctoral work developed commercial-microwave-link and Meteosat methods for rainfall monitoring in Sub-Saharan Africa.

Honors and Recognition

2026 AGU Open Science Recognition Prize – Global Precipitation Climatology Project Team. Member of the GPCP team recognized by the American Geophysical Union for long-term stewardship of open global precipitation data and trusted climate records. Official AGU recipient page

Professional Appointments

Postdoctoral Research Associate 2023–Present
Department of Hydrology & Atmospheric Sciences, University of Arizona, Tucson, AZ

- Develops and evaluates satellite precipitation retrievals and climate-data records, including AVHRR high-latitude retrieval and diagnostic evaluation for the published GPCP Version 3.3 record.
- Develops reusable algorithms, workflows, and toolsets for evaluating precipitation products, including GMI–GPROF, GPM IMERG, GPCP, ERA5, and MERRA-2 across global and regional scales, latitude and longitude bands, and surface states.
- Applies complementary reference datasets and a comprehensive evaluation framework—including distributional diagnostics (histograms and PDFs), categorical and continuous metrics, spatial diagnostics, and environmental stratification—to identify product strengths, biases, and retrieval sensitivities.

Visiting Scientist – EUMETSAT / H-SAF 2022–2023
European Organisation for the Exploitation of Meteorological Satellites, Darmstadt, Germany

Evaluated operational satellite precipitation products over Sub-Saharan Africa, with emphasis on diagnostic validation and uncertainty characterization.

Doctoral Researcher 2016–2022
Department of Water Resources, University of Twente, Enschede, Netherlands

Developed and published rainfall-detection and mapping methods combining commercial microwave-link attenuation with Meteosat Second Generation observations for data-sparse regions.

Education

Ph.D., GIS & Earth Observation – Hydrometeorology / Water Resources 2022
University of Twente, Faculty ITC, Enschede, Netherlands

Dissertation: *High-Spatiotemporal Resolution Rainfall Estimation from Satellite and Commercial Microwave Link Data.*

M.Sc., GIS & Earth Observation – Water Resources & Environmental Management 2016
University of Twente, Faculty ITC, Enschede, Netherlands

B.Sc., Environmental Science 2012
University of Cape Coast, Ghana

Selected Peer-Reviewed Publications and Open Dataset

Behrangi, A., Huffman, G. J., Adler, R. F., Song, Y., **Kumah, K. K.**, Zandi, O., Bolvin, D. T., Nelkin, E. J., & Gu, G. (2026). GPCP Version 3.3: A refined global view of precipitation from satellite and gauge observations. *Journal of Hydrometeorology*. Published online August 26, 2026. doi:10.1175/JHM-D-26-0048.1

Kumah, K. K., Zandi, O., & Behrangi, A. (2025). Retrospective mapping of global snow and ice cover beyond the satellite observational era. *Earth and Space Science*, 12(5), e2024EA004171. doi:10.1029/2024EA004171

Kumah, K. K., Maathuis, B. H. P., Hoedjes, J. C. B., & Su, Z. (2022). Near real-time estimation of high spatiotemporal resolution rainfall from cloud-top properties of MSG and CML rainfall intensities. *Atmospheric Research*, 279, 106357. doi:10.1016/j.atmosres.2022.106357

Kumah, K. K., Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2021). The MSG technique: Improving commercial microwave-link rainfall intensity using rain-area detection from Meteosat Second Generation. *Remote Sensing*, 13(16), 3274. doi:10.3390/rs13163274

Kumah, K. K., Maathuis, B. H. P., Hoedjes, J. C. B., Rwasoka, D. T., Retsios, B. V., & Su, B. Z. (2021). Rain-area detection in south-western Kenya using multispectral satellite data from Meteosat Second Generation. *Sensors*, 21(10), 3547. doi:10.3390/s21103547

David, N., Liu, Y., **Kumah, K. K.**, Hoedjes, J. C. B., Su, B. Z., & Gao, H. O. (2021). On the power of microwave communication data to monitor rain for agricultural needs in Africa. *Water*, 13(5), 730. doi:10.3390/w13050730

Kumah, K. K., Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2020). Combining MWL and MSG SEVIRI satellite signals for rainfall detection and estimation. *Atmosphere*, 11(9), 884. doi:10.3390/atmos11090884

Open dataset. Kumah, K. K., Zandi, O., & Behrangi, A. (2024). Global snow and ice cover (1980–1987): An extended GMASI dataset (Version 1) [Data set]. University of Arizona Research Data Repository. doi:10.25422/azu.data.28012532.v1

Preprints and Current Manuscripts

PREPRINT AVAILABLE – MANUSCRIPT SUBMITTED **Kumah, K. K.**, Behrangi, A., Zandi, O., Gardner, A. S., Wiese, D. N., & Greene, C. A. (2026). Quantifying Antarctic snowfall accumulation using the latest ice discharge and spaceborne gravity observations: Comparison with reanalysis and satellite precipitation products. *ESS Open Archive*. Preprint. doi:10.22541/essoar.15010201/v1.

SUBMITTED **Kumah, K. K.**, Behrangi, A., Huffman, G. J., Adler, R. F., Gu, G., Song, Y., Zandi, O., Bolvin, D. T., Nelkin, E. J., Funk, C. C., & Peterson, P. (2026). A multi-reference assessment of ocean precipitation products by integrating PAL, buoys, atolls, and OceanRAIN.

IN INTERNAL REVIEW **Kumah, K. K.**, Zandi, O., & Behrangi, A. (2026). Evaluating changes from GPROF Version 7 to Version 8 GMI precipitation retrievals over the contiguous United States.

Selected Conference Contributions

Behrangi, A., Huffman, G. J., Adler, R. F., Song, Y., **Kumah, K. K.**, Bolvin, D. T., Nelkin, E. J., & Gu, G. (2026). The newly released Global Precipitation Climatology Project (GPCP) V3.3 daily and monthly products and the future plans. *EGU General Assembly 2026*, Vienna, Austria, EGU26-5885. doi:10.5194/egusphere-egu26-5885

Zandi, O., Pfreundschuh, S., **Kumah, K. K.**, & Behrangi, A. (2025). University of Arizona High-latitude Infrared-based Precipitation Analysis Version 2 (UA-HIPA). *AGU Fall Meeting 2025*, New Orleans, Louisiana.

Kumah, K. K., Zandi, O., & Behrangi, A. (2024). On the retrospective mapping of global snow and ice cover beyond the satellite observational era. *AGU Fall Meeting 2024*, Washington, D.C.

Kumah, K. K. (2022). High spatial and temporal resolution rainfall estimation from MSG cloud-top properties using a random-forest algorithm trained with microwave-link-derived rainfall. *EUMETSAT Meteorological Satellite Conference 2022*, Brussels, Belgium, Paper 107, Session S07-B.

Kumah, K. K. (2019). Combining microwave links and Meteosat Second Generation data for rainfall studies in Kenya. *Symposium on the Hydrometeorological Usage of Data from Commercial Microwave Link Networks*, Garmisch-Partenkirchen, Germany.

Professional Service

Peer-review contributions. Completed manuscript reviews for *Journal of Geophysical Research: Atmospheres*, *Earth System Science Data*, and *Remote Sensing*, including an invited review for *JGR: Atmospheres* in 2025.

Proposal review. Completed an external proposal review for the Swiss National Science Foundation.

Research community service. Working Group 3 member, COST Action CA20136 SmartSense.

Selected Media and Research Features

Mobile Phone Signals Hold Real-Time Rainfall Information – APS Physics Magazine, 2021.

ITC algorithm detects rainfall in rural Africa – University of Twente / ITC, 2022.

ITC algorithm detects rainfall in rural Africa – myScience, 2022.

Researcher from UT developed an algorithm to predict rainfall in rural Africa – IO+, 2022.

A First for Ghana: Turning Mobile Phone Signals into Near-Real Time Rainfall Maps – TEMBO Africa, 2026.

Technical Methods

Scientific computing and data systems: Python, NumPy, Pandas, Xarray, Dask, scikit-learn, Git, Linux/HPC workflows; NetCDF, HDF5, GRIB, and GeoTIFF.

Remote sensing and geospatial analysis: AVHRR, MSG SEVIRI, GPM/IMERG, GPCP, GRACE/GRACE-FO, ERA5, MERRA-2, radar-gauge products, GIS and raster processing.

Retrieval and evaluation: machine learning, satellite precipitation retrieval, multi-reference validation, uncertainty diagnostics, climate-data records, bias assessment, and reproducible production pipelines.